As well-known, $(\mathbb{R}, +, \times)$ is the Complete ordered field.
Consider a set $\mathbb{R}^2$. Define two operations on $\mathbb{R}^2$ as:

$$(x_1, y_1) + (x_2, y_2) \overset{\text{def}}{=} (x_1 + x_2, y_1 + y_2)$$
$$(x_1, y_1) \cdot (x_2, y_2) \overset{\text{def}}{=} (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2)$$

Operation on $\mathbb{R}^2$

Defined on $\mathbb{R}$.

$(x, y)^{-1} =$

Then, the triple $(\mathbb{R}^2, +, \cdot)$ is a field:
$$\begin{cases} \text{Identity } (0,0) \\ \text{Unity } (1,0) \\ \text{inverse } (x,y)^{-1} = \left(\dfrac{x}{x^2+y^2}, \dfrac{-y}{x^2+y^2} \right) \end{cases}$$

Def. Define a Norm of $(a,b) \in \mathbb{C}$ as:
$$\|(a,b)\| \overset{\text{def}}{=} \sqrt{a^2+b^2}.$$

Lemma. $\|z\|^2 = z \cdot \bar{z}$. Moreover, if $z \neq 0$, then $z^{-1} = \dfrac{\bar{z}}{\|z\|^2}$

$(ih) \cdot (-ih)$
$= -i^2 \cdot h^2 = h^2.$

Consider a Complex map:
$$f : \Omega \to \mathbb{C} : (x,y) \mapsto (u(x,y), v(x,y))$$
Where $\Omega \subseteq \mathbb{C}$. Set Theoretically, it equals to a map from a subset of $\mathbb{R}^2$ into $\mathbb{R}^2$.
Moreover, Topologically, $\mathbb{C}$ is a Metric space induced by $\|\cdot\|$, homeomorphic to $\mathbb{R}^2$.
Therefore, the continuity of f is not different whether we see that $\mathbb{C}$ as $\mathbb{C}$ or $\mathbb{R}^2$.
But, in the Differentiation, Complex and $\mathbb{R}^2$ have crucial different. Recall that

$f : A (\subset \mathbb{R}^m) \to \mathbb{R}^n$ is differentiable at $a \in A^\circ$ if and only if; there exists a linear $T : \mathbb{R}^m \to \mathbb{R}^n$ s.t
$$\frac{f(a+h) - f(a) - T(h)}{\|h\|_m} \to 0 \quad \text{as} \quad h \to 0.$$
If the derivative T of f at $a \in A^\circ$ exists, it is unique.
Equivalently, if f has a derivative T at $a \in A^\circ$, then for small enough $h \in \mathbb{R}^m$,
$$f(a+h) - f(a) = \|h\| \cdot \left[\frac{f(a+h) - f(a) - T(h)}{\|h\|} \right] + T(h)$$
Moreover, if f is differentiable at $a \in A^\circ$, then all directional derivatives of f at a exist,
$$f'(a, u) = Df(a) \cdot u$$
But, since $\mathbb{C}$ is a field, we can define derivative without a norm,
this is the first bifurcation that indicates $\mathbb{C}$ does not equal $\mathbb{R}^2$.

「Def. Let $\Omega \subseteq \mathbb{C}$ open, and let $f : \Omega \to \mathbb{C}$ be a map.
Given $z_0 \in \Omega$, f is said to be Holomorphic at z_0 iff the limit

$$\lim_{h \to 0} \frac{f(z_0+h) - f(z_0)}{h} \quad (h \in \mathbb{C},\ h \neq 0)$$

exists. The limit is called a derivative of f at z_0, denote $f'(z_0)$.
Note: Holomorphic = Regular = Complex differentiable $\underset{\text{not definition.}}{\overset{}{=}}$ Analytic

「Cauchy - Riemann equation.
Let $f : \Omega \to \mathbb{C} : (x,y) \mapsto (u(x,y), v(x,y))$ be a map where $\Omega \subseteq \mathbb{C}$ open set.
If f is holomorphic at $z_0 = (x_0, y_0)$, then the partial derivatives of u, v at z_0 exist and

$$\frac{\partial u}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0) \quad \text{and} \quad \frac{\partial u}{\partial y}(z_0) = -\frac{\partial v}{\partial x}(z_0) \quad \cdots (*)$$

hold. (Or $u_x(z_0) = v_y(z_0)$ and $u_y(z_0) = -v_x(z_0)$)
Conversely, if u and v are continuously differentiable at $z_0 = (x_0, y_0)$ satisfying $(*)$, then f is holomorphic at z_0. More specifically,

$$f'(z_0) = \frac{\partial u}{\partial x}(z_0) + i \cdot \frac{\partial v}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0) - i \cdot \frac{\partial u}{\partial y}(z_0)$$

proof. Suppose that f is holomorphic at z_0. Then,

$$\begin{aligned} f'(z_0) &= \lim_{h_1 \to 0} \frac{f(x_0+h_1, y_0) - f(x_0, y_0)}{h_1} = \lim_{h_2 \to 0} \frac{f(x_0, y_0+h_2) - f(x_0, y_0)}{ih_2} \\ &= \qquad \frac{\partial f}{\partial x}(z_0) \qquad\qquad = \qquad \frac{1}{i} \cdot \frac{\partial f}{\partial y}(z_0) \end{aligned}$$

Therefore, at z_0, $u_x + i \cdot v_x = v_y - i \cdot u_y$.
Conversely, Suppose that u, v are continuously differentiable at z_0. That is, u_x, u_y, v_x, v_y exist and are continuous at z_0.

Thm. Define a Complex map as

$$f : \mathbb{D}_{<R} \to \mathbb{C} ; z \mapsto \sum_{n=0}^{\infty} a_n z^n$$

where $R = \left[\limsup_{n\to\infty} |a_n|^{\frac{1}{n}}\right]^{-1}$. Then, f is differentiable by terms on the domain as

$$f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}.$$

Proof. Since $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$, we obtain

$$\limsup |a_n|^{\frac{1}{n}} = \limsup |n a_n|^{\frac{1}{n}}$$

Thus $\sum a_n z^n$ and $\sum n a_n z^{n-1}$ have same Convergence boundary.
Define a map on $\mathbb{D}_{<R}$ as:

$$g(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$$

Enough to show that $g = f'$. Let $z_0 \in \mathbb{D}_{<R}$ be fixed, and $r \in \mathbb{R}$ s.t. $|z_0| < r < R$.
Since $f(z)$ converges, denote $S_N(z) = \sum_{n=0}^{N} a_n z^n$ and $E_N(z) = \sum_{n=N+1}^{\infty} a_n z^n$. ($f = S_N + E_N$).

For $h \in \mathbb{C}$ s.t. $|z_0 + h| < r$,

$$\frac{f(z_0+h) - f(z_0)}{h} - g(z_0) = \left[\frac{S_N(z_0+h) - S_N(z_0)}{h} - S_N'(z_0)\right] + \left(S_N'(z_0) - g(z_0)\right) + \left[\frac{E_N(z_0+h) - E_N(z_0)}{h}\right]$$

Meanwhile, $\left|\frac{E_N(z_0+h) - E_N(z_0)}{h}\right| \le \sum_{n=N+1}^{\infty} |a_n| \cdot \left|\frac{(z_0+h)^n - z_0^n}{h}\right| \le \sum_{n=N+1}^{\infty} |a_n| \cdot n \cdot r^{n-1} \to 0$ as $N \to \infty$

$$(z_0+h)^n - z_0^n = (z_0 + h - z_0)\left[(z_0+h)^{n-1} + (z_0+h)^{n-1} z_0 + \cdots + z_0^{n-1}\right]$$

And clearly $S_N'(z_0) - g(z_0) \to 0$ as $N \to \infty$. And,

$$\left|\frac{S_N(z_0+h) - S_N(z_0)}{h} - S_N'(z_0)\right| \to 0 \quad \text{as } h \to 0.$$

Therefore $g = f'$ for any $z_0 \in \mathbb{D}_{<R}$.

Corollary. Power Series defined on Convergence disk is C^{∞}.

⌜Curve.

A continuous map $z:[a,b]\to\mathbb{C}; t\mapsto z(t)$ is called a parametrized curve.
If it is injection, then called simple,
If $z(a)=z(b)$, then called closed.
It is called simple closed if injective on $[a,b)$.
Moreover, if z is continuously differentiable on $[a,b]$ and $z'\neq 0$, then it is smooth.

A smooth curves $\begin{cases} z:[a,b]\to\mathbb{C} \\ \tilde{z}:[c,d]\to\mathbb{C} \end{cases}$ are equivalent if there is $t:[c,d]\to[a,b]$ s.t

t is 1-1, t is continuously differentiable, $t'>0$, $\tilde{z}=z\circ t$.

「Integral.

「Def. Let γ be a smooth curve parametrized by $z: [a,b] \to \mathbb{C}$, and let $f: \mathbb{C} \to \mathbb{C}$ be a continuous map on γ. Define an integral of f along γ as;

$$\int_\gamma f(z)\, dz \overset{\text{def}}{=} \int_a^b f(z(t)) \cdot z'(t)\, dt.$$

And $\text{length}(\gamma) = \int_a^b |z'(t)|\, dt.$

「Thm. Let $f: \Omega \to \mathbb{C}$ be a map where $\Omega \subseteq \mathbb{C}$ is open, and γ be a curve from w_1 to w_2. If f has primitive function F on Ω, then

$$\int_\gamma f(z)\, dz = F(w_2) - F(w_1).$$

「Gaurount's Thm. Let $f: \Omega \to \mathbb{C}$ be a Holomorphic and a triangle $T \subseteq \Omega$. Then,

$$\int_T f(z)\, dz = 0.$$

「Lemma. In open disk, a Holomorphic map has a primitive map in the disk.

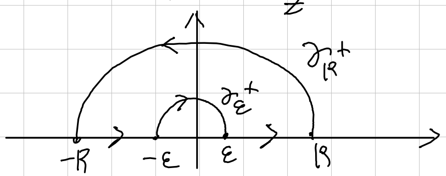
「Corollary. If f is Holomorphic in the disk, then for any closed curve γ in the disk,

$$\int_\gamma f(z)\, dz = 0.$$

「Calculate

Prove $\displaystyle\int_0^\infty \frac{1-\cos x}{x^2}\,dx = \frac{\pi}{2}$.

Prof. Let $f(z) = \dfrac{1-e^{iz}}{z^2}$. And, set the Countor as



observe that : for any $0<\varepsilon<R<\infty$,

$$\int_\gamma f(z)\,dz = \int_{-R}^{-\varepsilon} \frac{1-e^{ix}}{x^2}\,dx + \int_{\gamma_\varepsilon^+} \frac{1-e^{iz}}{z^2}\,dz + \int_\varepsilon^R \frac{1-e^{ix}}{x^2}\,dx + \int_{\gamma_R^+} \frac{1-e^{iz}}{z^2}\,dz = 0.$$

First, $\left|\dfrac{1-e^{iz}}{z^2}\right| \le \dfrac{2}{|z|^2}$ implies $\left|\displaystyle\int_\gamma f(z)\,dz\right| \le \sup_{z\in\gamma_R^+}\left|\dfrac{1}{z^2}\right| \cdot \text{length } \gamma_R^+ = \dfrac{1}{R^2}\cdot \pi R \to 0$

as $R\to\infty$. Therefore, $\displaystyle\int_{|x|\ge\varepsilon} \frac{1-e^{ix}}{x^2}\,dx = -\int_{\gamma_\varepsilon^+} \frac{1-e^{iz}}{z^2}\,dz.$

Since $\gamma_\varepsilon^- : z(t) = \varepsilon\cdot e^{it}$ $(t\in[0,\pi])$, and observe that

$$\frac{1-e^{iz}}{z^2} = \frac{-zi+zi+1-e^{iz}}{z^2} = -\frac{i}{z} + \frac{1+zi-e^{iz}}{z}$$

By L'Hopital Thm, $\displaystyle\lim_{z\to0} \frac{1+zi-e^{iz}}{z} = \lim_{z\to0} i+ie^{iz} = i+i = 2i.$

Therefore, $\dfrac{1+zi-e^{iz}}{z}$ is bounded in the neiborhood of zero. Take $M>0$ as bound.

Now, $\displaystyle\int_{\gamma_\varepsilon^-} -\frac{i}{z}\,dz = -\int_0^\pi \frac{i}{\varepsilon\cdot e^{it}}\cdot\varepsilon\cdot i\cdot e^{it}\,dt = \int_0^\pi 1\,dt = \pi$

and $\displaystyle\int_{\gamma_\varepsilon^-} \frac{1+zi-e^{iz}}{z}\,dz \le M\cdot\pi\varepsilon \to 0$ as $\varepsilon\to0$

Finally, $\displaystyle\int_{-\infty}^\infty \frac{1-\cos x}{x^2}\,dx = \lim_{\varepsilon\to0}\int_{|x|\ge\varepsilon} f(z)\,dz = \pi.$

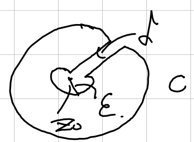
Since $\dfrac{1-\cos x}{x^2}$ is even, $\displaystyle\int_0^\infty \frac{1-\cos x}{x^2}\,dx = \frac{\pi}{2}.$

Thm. Cauchy Integral Formular.

suppose f is Holomorphic on some open set containing a closure of disk D.
If C is a curve with positive orientation of ∂D, then for any $z_0 \in D$,

$$f(z_0) = \frac{1}{2\pi i} \cdot \int_C \frac{f(z)}{z - z_0}\, dz$$

Proof. Let $z_0 \in D$ be fixed, and consider a curve $\Gamma_{\delta,\varepsilon}$ as



Since $\dfrac{f(z)}{z - z_0}$ is Holomorphic on $D \setminus \{z_0\}$, therefore

$\displaystyle\int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0}\, dz = 0$. Taking $\delta \to 0$, then we obtain

$$\lim_{\delta \to 0} \int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0}\, dz = \int_{C_\varepsilon} \frac{f(z)}{z - z_0}\, dz + \int_C \frac{f(z)}{z - z_0}\, dz = 0.$$

Meanwhile, $\dfrac{f(z)}{z - z_0} = \dfrac{f(z) - f(z_0)}{z - z_0} + \dfrac{f(z_0)}{z - z_0}$ and since f is Holomorphic,
the first term is bounded, therefore $\displaystyle\lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{f(z) - f(z_0)}{z - z_0}\, dz = 0$.

Now, $$\lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{f(z_0)}{z - z_0}\, dz = f(z_0) \cdot \lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{1}{z - z_0}\, dz$$
$$= f(z_0) \cdot \lim_{\varepsilon \to 0} \left(- \int_0^{2\pi} \frac{1}{\varepsilon \cdot e^{it}} \cdot \varepsilon \cdot i \cdot e^{it}\, dt \right)$$
$$= - f(z_0) \cdot 2\pi i.$$

Consequently, $\displaystyle\int_C \frac{f(z)}{z - z_0}\, dz = 2\pi i\, f(z_0)$.

Corollary. $\displaystyle\int_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz = 2\pi i \cdot \frac{f^{(n)}(z_0)}{n!}$.

Proof. Using induction for n. For $n=0$, clear. If $f^{(n-1)}(z_0) = \dfrac{(n-1)!}{2\pi} \cdot \displaystyle\int_C \frac{f(z)}{(z - z_0)^n}\, dz$, then

$$\frac{f^{(n-1)}(z_0 + h) - f^{(n-1)}(z_0)}{h} = \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \left[\frac{1}{(z - z_0 - h)^n} - \frac{1}{(z - z_0)^n} \right] dz$$
$$= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \cdot \left[\frac{h}{(z - z_0 - h)(z - z_0)} \cdot \left(\sum_{k=0}^{n-1} \frac{1}{(z - z_0 - h)^{n-1-k}} \cdot \frac{1}{(z - z_0)^k} \right) \right] dz$$
$$\to \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{(z - z_0)^2} \cdot \sum_{k=0}^{n-1} \frac{1}{(z - z_0)^{n-1}}\, dz \quad \text{as } h \to 0$$
$$= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot n \cdot \frac{1}{(z - z_0)^{n+1}}\, dz$$
$$= \frac{n!}{2\pi i} \cdot \int_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz = f^{(n)}(z_0).$$

Thm. Cauchy inequality.

$|f^{(n)}(z_0)| \le \dfrac{n!}{R} \cdot \sup_C |f|$ if f is Holomorphic in an open containing $\overline{B_R(z_0)}$, $C = \partial B_R(z_0)$.

Cauchy's Integral Formula, Revised.

Thm. Let $f: G \to \mathbb{C}$ be a Holomorphic on open $G \subset \mathbb{C}$.
If $z_0 \in G$ and $\overline{B_r(z_0)} \subseteq G$, then for a Curve $C: z_0 + re^{it}$ $(0 \le t \le 2\pi)$,

$$f(z_0) = \frac{1}{2\pi i} \cdot \oint_C \frac{f(z)}{z - z_0}\, dz.$$

Prf. Denote $D = B_r(z_0)$. Let $z_0 \in D$ be fixed, and consider a Curve $\Gamma_{\delta,\varepsilon}$ as

Since $\dfrac{f(z)}{z - z_0}$ is Holomorphic on $D \setminus \{z_0\}$, $\displaystyle\int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0}\, dz = 0$.

Taking $\delta \to 0$, then we obtain

$$\lim_{\delta \to 0} \int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0}\, dz = -\int_{C_\varepsilon} \frac{f(z)}{z - z_0}\, dz + \int_C \frac{f(z)}{z - z_0}\, dz = 0.$$

Meanwhile, $\dfrac{f(z)}{z - z_0} = \dfrac{f(z) - f(z_0)}{z - z_0} + \dfrac{f(z_0)}{z - z_0}$ and since f is Holomorphic,
the first term is bounded, therefore $\displaystyle\lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{f(z) - f(z_0)}{z - z_0}\, dz = 0$.

Now,
$$\lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{f(z_0)}{z - z_0}\, dz = f(z_0) \cdot \lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{1}{z - z_0}\, dz$$
$$= f(z_0) \cdot \lim_{\varepsilon \to 0} \left(\int_0^{2\pi} \frac{1}{\varepsilon \cdot e^{it}} \cdot \varepsilon \cdot i \cdot e^{it}\, dt \right)$$
$$= f(z_0) \cdot 2\pi i.$$

$$\int_C \frac{f(z)}{z - z_0}\, dz = 2\pi i\, f(z_0).$$

Consequently,

$$\int_C \frac{f(z)}{z - z_0}\, dz = \lim_{\delta \to 0} \int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0}\, dz + \int_{C_\varepsilon} \frac{f(z)}{z - z_0}\, dz = 2\pi i \cdot f(z_0).$$

Corollary. $\displaystyle\int_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz = 2\pi i \cdot \frac{f^{(n)}(z_0)}{n!}$

「Thm. Suppose f is Holomorphic on an open G containing a closed disk $\hat{D}$ having center z_0,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n \quad (z \in D)$$

Proof. $f(z) = \frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{\mu - z} \, d\mu \quad \left(\frac{1}{\mu - z} = \frac{1}{\mu - z_0 - (z - z_0)} = \frac{1}{\mu - z_0} \cdot \frac{1}{1 - \left(\frac{z - z_0}{\mu - z_0}\right)} \right)$

$$= \frac{1}{2\pi i} \cdot \int_C f(\mu) \cdot \sum_{n=0}^{\infty} \frac{1}{(\mu - z_0)^{n+1}} \cdot (z - z_0)^n \, d\mu$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{(\mu - z_0)^{n+1}} \, d\mu \right) \cdot (z - z_0)^n$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n.$$

「Corollary, Liouville's Theorem.

If $f : \mathbb{C} \to \mathbb{C}$ is entire and bounded, then it is constant.

Proof. Given $z_0 \in \mathbb{C}$ and $R > 0$, $|f'(z_0)| \le \frac{1}{R} \cdot \sup_{C_R} |f| \to 0$ as $R \to \infty$. Therefore, $f' = 0$

Thus f is Constant.

「Fundamental Theorem of Algebra.

Non-constant polynomial $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0 \in \mathbb{C}[z]$ has a root in $\mathbb{C}$.

Proof. Suppose that $P(z)$ has no root in $\mathbb{C}$. Then, $1/P(z)$ is well-defined on $\mathbb{C}$, entire.

observe that $\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n} \right) \quad (z \neq 0)$.

The terms $a_{n-k} \cdot z^{-k}$ $(k = 1, \cdots, n)$ Converges to zero as $|z| \to \infty$, thus for some $R > 0$,

$$|P(z)| \ge 2|a_n| \cdot z^n \quad (|z| > R)$$
$$\ge 2|a_n| \cdot R^n$$

Therefore it is bounded below in $|z| > R$, and clearly in $|z| \le R$.

Now, $\frac{1}{P}$ is bounded, thus Constant. Contradiction.

Thm. Let G be a Connected open set and let $f: G \to \mathbb{C}$ be an analytic function
Then, TFAE:

a) $f \equiv 0$.

b) There is a point $a \in G$ such that $f^{(n)}(a) = 0$ for each $n \geq 0$.

c). $f^{-1}[\{0\}]$ has a limit point in G.

Proof. a) $\Rightarrow$ b) and a) $\Rightarrow$ c) are trivial. To show c) $\Rightarrow$ b),
Let $a \in G$ be an limit point of $Z = f^{-1}[\{0\}]$. Since Z is closed, $a \in Z$, or $f(a) = 0$.
Suppose that for some $n \in \mathbb{N}$,

$$f(a) = f'(a) = \cdots = f^{(n-1)}(a) = 0 \quad \text{but } f^{(n)}(a) \neq 0.$$

Since f is analytic, $f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} \cdot (z-a)^k = \sum_{k=n}^{\infty} \frac{f^{(k)}(a)}{k!} \cdot (z-a)^k$.

in $|z-a| < R$, where $R > 0$ s.t $B_R(a) \subset G$.
Put $g(z) = \sum_{k=n}^{\infty} a_k (z-a)^{k-n} \quad \left(a_k = \frac{f^{(k)}(a)}{k!}\right)$

then, g is analytic in $B_R(a)$, and $f = (z-a)^n g$, and $g(a) = a_n \neq 0$.
Now, we can find $0 < r < R$ s.t $g(z) \neq 0$ for $|z-a| < r$, but since a is the limit point of Z, so for some b s.t $f(b) = 0$ and $0 < |b-a| < r$, thus

$$f(b) = 0 = (b-a)^n \cdot g(b) \Rightarrow g(b) = 0,$$

which is Contradiction.

Singularities.

Def. A complex function f has an isolated singularity at $z=a$ if: there is an $R>0$ s.t. f is defined and analytic in $B_R(a)-\{a\}$, but not in $B_R(a)$.

An isolated singularity $a\in\mathbb{C}$ of f is called:

- Removable if: there is an analytic extension of f to $B_R(a)$.
- Pole if: $\lim_{z\to a}|f(z)|=\infty$.
- essential if: neither Removable nor Pole.

Thm. Let $a\in\mathbb{C}$ be an isolated singularity of f. Then,

$a\in\mathbb{C}$ is removable singularity iff $\lim_{z\to a}(z-a)\cdot f(z)=0$.

Prop. If $G\subseteq\mathbb{C}$ is a region containing $a\in\mathbb{C}$ and f is analytic on $G\setminus\{a\}$ with a pole a, then there exists $m\in\mathbb{N}$ and analytic $g:G\to\mathbb{C}$ such that

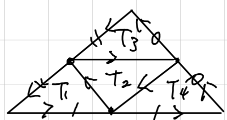
$$f(z)=\frac{g(z)}{(z-a)^m}$$

The smallest such a number is called order of a pole.

「Cauchy - Goursat's Thm.

「Thm. A map f is Holomorphic on open set $\Omega \subseteq \mathbb{C}$ and a triangle T is contained in Ω then $\int_T f(z)\,dz = 0$.

Proof. Let $T^{(0)} = T$ with positive oriented, and $T_1^{(1)}, T_2^{(1)}, T_3^{(1)}, T_4^{(1)}$ be;



Then, $\displaystyle\int_{T^{(0)}} f(z)\,dz = \sum_{i=1}^{4} \int_{T_i^{(1)}} f(z)\,dz$, because the integral on T_2 is canceled.

So, for some $j = 1, 2, 3, 4$, $\displaystyle\left|\int_{T^{(0)}} f(z)\,dz\right| \leq 4 \cdot \left|\int_{T_j^{(1)}} f(z)\,dz\right|$

Put $T^{(1)} = T_j^{(1)}$. Inductively, for each $n \in \mathbb{N}$,

$$\left|\int_{T^{(0)}} f(z)\,dz\right| \leq 4^n \cdot \left|\int_{T^{(n)}} f(z)\,dz\right|$$

And, by construction,

$$T^{(0)} \supseteq T^{(1)} \supseteq \cdots$$

Def. Let G be an open set of $\mathbb{C}$.
A function $f: G \to \mathbb{C}$ is called Analytic if f is continuously differentiable.

Logarithm

Def. If $G \subseteq \mathbb{C}$ is an open connected set in $\mathbb{C}$ and $f: G \to \mathbb{C}$ is a continuous function s.t

$$z = \exp f(z) \text{ for all } z \in G$$

Then, f is called a branch of the logarithm.

Explicitly, if $z = x+iy \in G$, then let $\log z = a+ib$. Now,

$$\exp(\log z) = \exp(a)\cdot\exp(ib) = e^a\cdot(\cos b + i\sin b) = x+iy.$$

Thus, $\begin{cases} e^a \cdot \cos b = x \\ e^a \cdot \sin b = y \end{cases}$, $|e^a \cdot (\cos b + i\sin b)| = |e^a| = |z|$, so $a = \ln|z|$.
$b = \arg z$.

Therefore, $\log z = \ln|z| + i\cdot\arg z$.

「Cauchy - Riemann equation.

Let $f: G \to \mathbb{C}$ be a map on region G and let $u(x,y) = \mathrm{Re}\, f(x,y)$, $v(x,y) = \mathrm{Im}\, f(x,y)$.
Suppose that u and v have continuous partial derivatives. Then,

f is analytic if and only if u and v satisfy the Cauchy-Riemann equation.

Proof suppose that f is analytic. For $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+h)-f(z)}{h} = \frac{u(x+h,y)-u(x,y)}{h} + i \cdot \frac{v(x+h,y)-v(x,y)}{h}$$

Letting $h \to 0$, then $f'(z) = \frac{\partial u}{\partial x}(x,y) + i \cdot \frac{\partial v}{\partial x}(x,y)$.

Meanwhile, for $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+ih)-f(z)}{ih} = -i \cdot \frac{u(x,y+h)-u(x,y)}{h} + i \cdot \frac{v(x,y+h)-v(x,y)}{ih}$$

Letting $h \to 0$, then $f'(z) = -i \cdot \frac{\partial u}{\partial y}(x,y) + \frac{\partial v}{\partial y}(x,y)$

So, $u_x = v_y$ and $v_x = -u_y$. $\cdots (*)$ not using continuously differentiable.

Conversely, u and v satisfy the condition $(*)$, and u_x, u_y, v_x, v_y are continuous.
Let $z = x+iy \in G$ be given. Since G is open, for some $r>0$, $B_r(z) \subseteq G$.
For $h = s+it \in B_r(0)$,

$$u(x+s,y+t) - u(x,y) = [u(x+s,y+t) - u(x,y+t)] + [u(x,y+t) - u(x,y)]$$

By the Mean-Value Theorem for variables s and t, respectively,

$u(x+s,y+t) - u(x,y+t) = u_x(x+s_1, y+t) \cdot s$ for some $|s_1| < |s|$

and

$u(x,y+t) - u(x,y) = u_y(x, y+t_1) \cdot t$ for some $|t_1| < |t|$ $\quad (*)$

Let $\varphi(s,t) = [u(x+s,y+t) - u(x,y)] - [u_x(x,y) \cdot s + u_y(x,y) \cdot t]$. Now, $(*)$ gives

$$\frac{\varphi(s,t)}{s+it} = \frac{s}{s+it} \cdot [u_x(x+s_1,y+t) - u_x(x,y)] + \frac{t}{s+it} \cdot [u_y(x,y+t_1) - u_y(x,y)]$$

But, $|s_1| < |s| \le |s+it|$ and $|t_1| < |t| \le |s+it|$, so letting $s+it \to 0$, then

$$\frac{\varphi(s,t)}{s+it} \to 0$$

Therefore, $u(x+s,y+t) - u(x,y) = u_x(x,y)s + u_y(x,y)t + \varphi(s,t)$

and similarly $v(x+s,y+t) - v(x,y) = v_x(x,y)s + v_y(x,y)t + \psi(s,t)$ where $\lim_{s+it \to 0} \frac{\psi(s,t)}{s+it} = 0$.

Now, $$\frac{f(z+s+it) - f(z)}{s+it} = u_x(z) + i v_x(z) + \frac{\varphi(s,t) + i\psi(s,t)}{s+it} \to u_x(z) + i v_x(z)$$

as $s+it \to 0$.

Thm. Let G be either $\mathbb{C}$ or some open disk
If $u: G \to \mathbb{R}$ is a harmonic, then u has a harmonic conjugate.
That is, there is $v: G \to \mathbb{R}$ s.t $u + iv$ is analytic in G.
Proof. Let $G = B_R(0)$, $0 < R \leq \infty$. For a harmonic $u: G \to \mathbb{R}$, define

$$v(x,y) = \int_0^y u_x(x,t)\,dt + \varphi(x).$$

where $\varphi(x)$ s.t $v_x = -u_y$. Now, differentiating both sides with respect to x:

$$\begin{aligned} v_x(x,y) &= \int_0^y u_{xx}(x,t)\,dt + \varphi'(x) \\ &= \int_0^y -u_{yy}(x,t)\,dt + \varphi'(x) \\ &= -u_y(x,y) + u_y(x,0) + \varphi'(x) \end{aligned}$$

So, $\varphi'(x) = -u_y(x,0)$. Now,

$$v(x,y) = \int_0^y u_x(x,t)\,dt - \int_0^x u_y(s,0)\,ds$$

satisfies the Cauchy-Riemann equation.

Zeros of Analytic functions.

Def. If $f: G \to \mathbb{C}$ is analytic and $a \in G$ s.t $f(a)=0$,

then a is called a zero of f of multiplicity $m \geq 1$ if:

there exists an analytic $g: G \to \mathbb{C}$ s.t $f(z)=(z-a)^m \cdot g(z)$, where $g(a) \neq 0$.

「Index of a Closed Curve.

「Let $\gamma(t) = a + e^{2\pi i n t}$ $(0 \le t \le 1)$. Then,

$$\int_\gamma \frac{1}{z-a}\, dz = \int_0^1 e^{-2\pi i n t} \cdot 2\pi i n \cdot e^{2\pi i n t}\, dt = 2n\pi \cdot i.$$

In general,

「Lemma. Let $\gamma : [0,1] \to \mathbb{C}$ be a closed rectifiable curve and $a \notin \operatorname{Im}\gamma$, then

$$\frac{1}{2\pi i} \cdot \int_\gamma \frac{1}{z-a}\, dz$$

is an integer.

Proof. Assume that γ is a smooth map. Define $g : [0,1] \to \mathbb{C}$ as

$$g(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s)-a}\, ds$$

Then, $g(0) = 0$ and $g(1) = \int_0^1 \frac{\gamma'(s)}{\gamma(s)-a}\, ds = \int_\gamma \frac{1}{z-a}\, dz$

And $g'(t) = \dfrac{\gamma'(t)}{\gamma(t)-a}$ for $0 \le t \le 1$.

But, $\dfrac{d}{dt} e^{-g(t)} \cdot (\gamma(t)-a) = e^{-g(t)} \cdot \gamma'(t) - g'(t) \cdot e^{-g(t)} \cdot (\gamma(t)-a) = 0$

So, $e^{-g(t)}(\gamma(t)-a)$ is the constant,

$$e^{-g(0)}(\gamma(0)-a) = \gamma(0) - a = e^{-g(1)}(\gamma(1)-a)$$

Since γ is closed curve, so $\gamma(1) = \gamma(0)$,

$$e^{-g(1)} = 1 \quad \text{or} \quad g(1) = 2k\pi \cdot i \text{ for some } k \in \mathbb{Z}.$$

「Def. If γ is a closed rectifiable curve in $\mathbb{C}$ and $a \notin \operatorname{Im}\gamma$,

$$n(\gamma; a) \overset{\text{def}}{=} \frac{1}{2\pi i} \cdot \int_\gamma \frac{1}{z-a}\, dz$$

is called the index of γ with respect to a

「Morera Thm,

「Thm. Let $f: G \to \mathbb{C}$ be a Continuous function on a region G Such that

$$\int_T f = 0$$

for every triangular path T in G, then f is analytic in G.

Proof. Enough to Show that f has a Primitive. WLOG, Let $G = B_R(a)$. Define

$$F(z) = \int_{[a,z]} f(w)\,dw, \quad \text{where } [a,z]: (1-t)a + tz, \quad 0 \le t \le 1.$$

Let $z_0 \in G$ be fixed. Then, $F(z) = \int_{[a,z_0]} f + \int_{[z_0,z]} f$, so

$$\frac{F(z) - F(z_0)}{z - z_0} = \frac{1}{z - z_0} \cdot \int_{[z_0,z]} f$$

Therefore, $\frac{F(z) - F(z_0)}{z - z_0} - f(z_0) = \frac{1}{z - z_0} \cdot \int_{[z_0,z]} f - f(z_0)$

$$= \frac{1}{z - z_0} \cdot \int_{[z_0,z]} [f(w) - f(z_0)]\,dw.$$

Now, $\left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| = \left| \frac{1}{z - z_0} \cdot \int_{[z_0,z]} [f(w) - f(z_0)]\,dw \right| \le |f(z) - f(z_0)| \to 0.$